LumiPath: An Adaptive Pedestrian-Safety Streetlight System Using Multi-Sensor Fusion for Smart Cities

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***Abstract*—Conventional streetlights burn all night whether or not anyone is present, wasting energy while still giving no help to a pedestrian in trouble. This paper presents LumiPath, a low-cost adaptive streetlight prototype built on an Arduino Uno. A light-dependent resistor (LDR), two infrared (IR) proximity sensors and an HC-SR04 ultrasonic sensor are combined so that a three-LED lamp turns on only when it is dark and someone is present, and turns off in the same control cycle once the person leaves. A push-button SOS toggles a buzzer, and live readings stream to a web dashboard. The work follows a software-engineering life cycle (SRS, SDS, implementation, testing, outcome) and is mapped to Smart Cities Mission themes and to UN Sustainable Development Goals 3, 7, 9, 11 and 13. A two-threshold hysteresis (900/850 ADC counts) with a motion-gated mode update keeps the day/night decision stable and immune to the lamp's own light. All seven bench test cases and the dashboard check passed. An analytical model, not a field measurement, estimates a 49% to 89% yearly energy saving per pole for a 70 W lamp depending on how often the road is occupied.**

***Index Terms*—smart city, smart street lighting, Arduino, IoT, pedestrian safety, hysteresis, sustainable development goals.**

I. INTRODUCTION

Street lighting is one of the most visible and most repeated pieces of city infrastructure. Most installations still run on a timer or a simple dusk-to-dawn photocell, so a lamp stays at full power on an empty road at 3 AM, yet may still be badly placed for someone crossing at 9 PM [1], [2]. A smart city should instead light the road in response to what is actually happening on it.

This paper describes LumiPath, a prototype streetlight that (i) stays off in daylight, (ii) stays off at night until presence is detected, (iii) switches off immediately when presence ends, (iv) measures distance and approximate speed of the approaching object, and (v) lets a person raise an alarm with a button. The prototype is deliberately cheap and built from common parts so that the same node could be repeated on every pole of a city.

The contributions of this paper are:

- a requirement-driven design (SRS and SDS) for a presence-aware streetlight node with seven functional and five non-functional requirements;
- a working Arduino implementation with a web dashboard, tested against requirement-traced cases;
- a two-threshold, motion-gated day/night controller that avoids both threshold flicker and self-illumination errors; and
- a mapping of the project to Smart Cities Mission themes and SDG metrics, with a transparent energy and carbon estimate.

The rest of the paper is organised as follows. Section II reviews related work, Section III shows how smart city concepts are applied, Sections IV to VII cover the SRS, SDS, implementation and testing, Section VIII presents the outcome and the unique results, and Section IX concludes.

II. RELATED WORK

Recent surveys agree that traditional lighting based on timers or simple photosensors is inefficient and that IoT-based control can adjust illumination from real-time data [2]. Omar *et al.* review intelligent street lighting systems that use sensors for light control and connectivity for weather and fault reporting, and compare the limitations of each [1]. Agramelal *et al.* classify smart streetlight control methods and report the energy-saving potential of simulated, prototype and in-field systems [3]. These works motivate sensor-driven control but are mostly reviews.

LumiPath is a small prototype that places the focus on four practical points on one node: presence-gated switching, a flicker-free day/night decision, distance and speed sensing, and a personal-safety SOS, all visible on a live dashboard. We use the Smart Cities Mission [4] and the 2030 Agenda [5] as the framework for judging its value.

III. SMART CITY CONCEPTS APPLIED

The Smart Cities Mission combines area-based development with pan-city solutions [4]. Smart street lighting is a typical pan-city solution because an identical unit is repeated on every pole. Table I links the main smart city concepts to what LumiPath does.

TABLE I
SMART CITY CONCEPTS AND THEIR REALISATION

Concept	Realisation in LumiPath
Smart energy	Lamp is off unless dark and presence is detected
Smart safety	Light on pedestrian presence; SOS buzzer
Smart infrastructure	Ordinary pole becomes a sensing node (5 inputs, 4 outputs)
IoT and ICT	Sense, decide and report with no operator; serial feed to dashboard

Concept	Realisation in LumiPath
Smart mobility	Distance and speed of approaching object from HC-SR04
Smart environment	Lower electricity use and CO_2 (Section VIII)

The project also supports five UN Sustainable Development Goals [5]. SDG 7 (target 7.3, energy efficiency) is measured in kWh saved per pole per year. SDG 9 (target 9.1, resilient infrastructure) is measured by the number of sensing functions added to existing poles and the response time. SDG 11 (target 11.7, safe public spaces) is measured by time from detection to light and by SOS availability. SDG 3 (target 3.6, road traffic injuries) is supported by better visibility where a pedestrian stands. SDG 13 (target 13.2) is measured in kg CO_2 avoided per pole per year.

IV. SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Requirements were written in the style recommended by ISO/IEC/IEEE 29148 [6]: each requirement is short, testable and has an identifier so that tests can be traced back to it. The system context is a single pole-mounted node that senses its surroundings and drives a lamp, a buzzer and a serial link. Table II lists the requirements.

TABLE II
FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

ID	Requirement
FR1	Read ambient light on A0; enter night above 900 and return to day only below 850.
FR2	Detect presence when either IR sensor output is LOW.
FR3	In night mode, switch all three LEDs ON while presence exists and OFF as soon as it ends.
FR4	In day mode, keep all LEDs OFF regardless of presence.
FR5	Measure distance with the HC-SR04 and compute speed from successive readings.
FR6	Toggle the buzzer ON/OFF on each SOS button press, in any mode.
FR7	Send LDR value, IR states, distance and speed over serial at 9600 baud.
NFR1	Respond to a change in presence within one control loop (about 0.2 s).
NFR2	No day/night flicker near the threshold (50-count hysteresis band).
NFR3	The lamp's own light must not change the day/night decision.
NFR4	Use low-cost commodity parts and the Arduino IDE.
NFR5	Ignore button bounce (200 ms hold-off after a press).

V. SOFTWARE DESIGN SPECIFICATION (SDS)

The design follows the viewpoints of IEEE Std 1016 [7]: architecture, interface, and behaviour.

A. Architecture

The node has three layers (Fig. 1). The sensing layer holds the LDR, two IR modules, the HC-SR04 and the SOS button. The control layer is an Arduino Uno [8] that runs one non-blocking-style loop. The actuation and reporting layer holds three LEDs, a buzzer and the serial port feeding the dashboard.

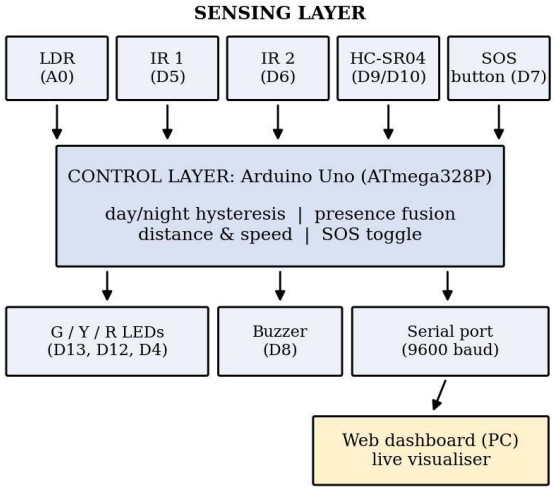


Fig. 1. Three-layer architecture of the LumiPath node.

B. Interface design

Table III gives the pin assignment. IR inputs are active-LOW, the SOS input uses the internal pull-up, and the HC-SR04 uses a trigger/echo pair [9].

TABLE III
PIN ASSIGNMENT

Device	Pin	Device	Pin
Green LED	D13	Ultrasonic trig	D9
Yellow LED	D12	Ultrasonic echo	D10
Red LED	D4	LDR	A0
IR sensor 1	D5	Buzzer	D8
IR sensor 2	D6	SOS button	D7

C. Behavioural design

Behaviour is a three-state machine (Fig. 2). Day/night is updated only in states where no presence is detected. This single rule is what makes the controller immune to the lamp's own light, because the LEDs are on only while presence exists.

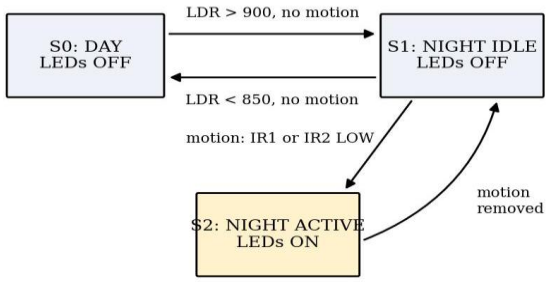


Fig. 2. State model: day, night idle and night active.

Distance and speed use (1) and (2), where t is the echo time in µs, d_k is the current distance in cm and Δt is the time between readings in s. The speed obtained is the radial speed along the sensor axis.

$$d = (t \times 0.034) / 2 \quad (1)$$

$$v = | d_k - d_{k-1} | / \Delta t \quad (2)$$

VI. IMPLEMENTATION

The hardware was built on a breadboard around an Arduino Uno (Fig. 3). The software is a single Arduino C++ sketch with one control loop of about 200 ms and a helper function that drives all three LEDs together. Each loop does the following: read the LDR and both IR inputs; update the day/night mode only if no presence is found; check the SOS button edge and toggle the buzzer; trigger the HC-SR04 and compute distance and speed; print one serial line; and set the LEDs.

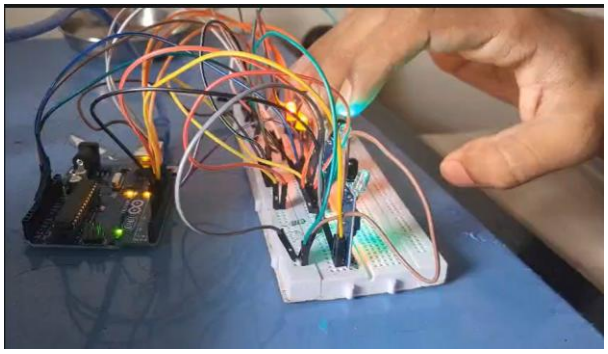


Fig. 3. Prototype on a breadboard, LEDs lit by a hand near the IR sensors.

A web dashboard, the Adaptive Streetlight Interactive Visualizer, shows a road with a pole and light cone, the LDR bar with both thresholds, the state of both IR sensors, distance, speed and the three LED arms (Fig. 4). During the demonstration it ran on a laptop beside the circuit and showed readings of about 30 cm distance and a few cm/s speed.



Fig. 4. Dashboard running beside the hardware.

VII. TESTING

Testing was done on the bench by changing the light level over the LDR, moving a hand in front of the IR and ultrasonic sensors and pressing the button. Each test is traced to a requirement in Table IV. Results follow the team's bench observations. NFR2 and NFR3 were met by design (Fig. 5) and confirmed during threshold tuning.

TABLE IV
TEST CASES TRACED TO REQUIREMENTS

ID	Test and expected result	Req.	Result
T1	Bright light, hand near IR: LEDs stay OFF	FR4	Pass
T2	Dark, no presence: LEDs OFF	FR3	Pass
T3	Dark, hand near IR: all LEDs ON at once	FR2, FR3	Pass
T4	Dark, hand removed: LEDs OFF at once	FR3, NFR1	Pass
T5	Hand toward HC-SR04: distance falls, speed changes	FR5	Pass
T6	SOS pressed once: buzzer ON	FR6	Pass
T7	SOS pressed again: buzzer OFF	FR6, NFR5	Pass
T8	Readings visible on dashboard	FR7	Pass

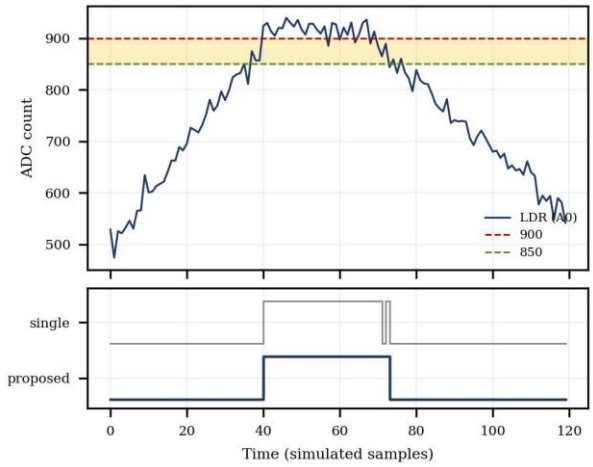


Fig. 5. Two-threshold switching versus a single threshold (simulated signal).

VIII. OUTCOME AND UNIQUE RESULTS

A. Project outcome

The prototype met all seven functional requirements in bench tests. The lamp stayed off in daylight and in an empty dark scene, lit within one loop of detection and went off within one loop of removal. The SOS worked in both modes and the dashboard showed all sensor values. Table V lists the measured and design values.

TABLE V
PROTOTYPE METRICS

Metric	Value
Day/night thresholds	900 (night), 850 (day)
Hysteresis band	50 counts (about 4.9% of 1023)
Control loop period	200 ms
LED response to presence	Within one loop (about 0.2 s)
Sensing inputs / outputs	5 / 4 (3 LEDs, 1 buzzer) + serial

B. Estimated energy and carbon impact

No lamp power was measured on the breadboard, so an analytical model is used. Let P be lamp power, H the night length in hours, N the number of nights, α the share of night with presence, and P^c the controller power. Then (3) gives the yearly energy and (4) the saving.

$$E = P \cdot H \cdot N \cdot \alpha + P^c \cdot H \cdot N \quad (3)$$

$$S = 1 - E / (P \cdot H \cdot N) \quad (4)$$

We assume $P = 70$ W, $H = 12$ h, $N = 365$, $P^c = 0.5$ W and a grid factor of 0.8 kg CO_2/kWh. A conventional lamp then uses about 307 kWh/yr. Table VI and Fig. 6 give the results. These values are estimates and ignore dimming and lamp-driver losses.

TABLE VI
ESTIMATED YEARLY ENERGY PER POLE

Presence	kWh/yr	Saved	S	t CO_2, 10k poles
10% of night	33	274	89%	2,190
20% of night	64	243	79%	1,945
30% of night	94	212	69%	1,699
50% of night	155	151	49%	1,209

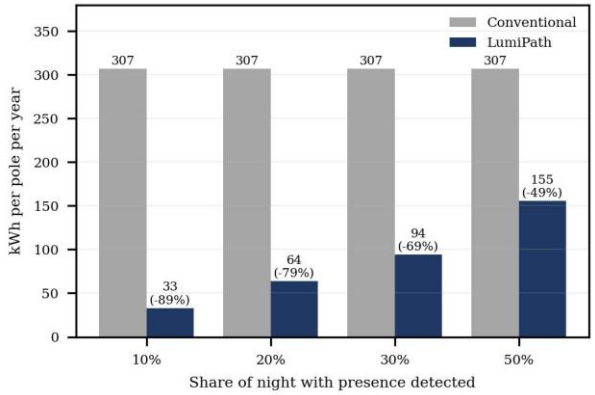


Fig. 6. Estimated yearly energy per pole, conventional versus LumiPath.

C. Unique outcomes

Four results go beyond a basic motion-activated light:

- **Self-illumination-proof day/night logic.** The mode is updated only when no presence exists, so the lamp's own light on the LDR cannot cause a false switch to day. This removes a common failure of LDR-controlled lights that are placed near their own lamp.
- **Flicker-free mode with a 50-count band.** Two thresholds replace one, so a noisy reading near the limit changes the mode only once (Fig. 5).
- **Safety and sensing on one node.** Presence lighting, distance and speed sensing and a toggle SOS share one low-cost controller, rather than being separate devices.
- **SDG-linked evidence.** The report ties each SDG to a measurable quantity (kWh, kg CO_2, response time) and states clearly which numbers are estimates, so the outcome can be checked later by a field trial.

IX. LIMITATIONS AND FUTURE WORK

The prototype uses LEDs instead of a real lamp, which would need a relay or driver. IR sensors have short range and suffer in strong sunlight. Speed is only displayed and not yet used for control. The mode is frozen while presence is detected, so a dawn that arrives during continuous presence is noticed only after the area is clear. The ultrasonic call waits for an echo without a time-out, so a missing echo can stall one loop. The SOS sounds a local buzzer only. Future work includes a Wi-Fi link (ESP8266 or ESP32) to monitor many poles, PWM dimming instead of ON/OFF, speed-based pedestrian and vehicle classification, an SOS notification to a contact, solar supply, and a field trial with a power meter to replace the estimates in Section VIII.

X. CONCLUSION

LumiPath shows that a streetlight which responds to darkness and presence can be built from a few cheap parts. Following an SRS, SDS, implementation and testing path gave a prototype that met all its functional requirements on the bench, and the two-threshold, motion-gated mode update gave a stable, self-illumination-proof controller. The estimated energy saving of roughly half to nearly nine-tenths per pole, together with the safety features, links the work directly to the Smart Cities Mission and to SDGs 3, 7, 9, 11 and 13. A field trial is the next step to confirm the estimates.

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